

Mirror Neurons

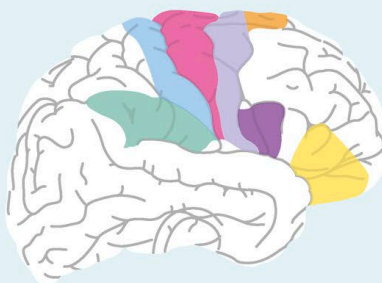
Learning does not just involve practicing a new skill—we also learn by watching others. This kind of learning is thought to involve nerve cells in the brain called mirror neurons that allow us to experience actions without actually performing them.

What are mirror neurons?

Mirror neurons are brain cells that fire both when we perform an action and when we see someone else performing that action. They were first discovered in monkeys but have since been found in humans, too. Most studies have used functional magnetic resonance imaging (fMRI; see p.43), but one study involved people who had electrodes implanted into their brains. In this instance, mirror neuron cells were detected in the supplementary motor area, where movement sequences are planned, as well as in the hippocampus, which governs memory and navigation.

Where are they?

Mirror neurons have been found in several cortical areas as well as in structures deeper within the brain, such as the hippocampus.



KEY

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|--------------------------|------------------------|
| Premotor area | Primary motor area |
| Part of Broca's area | Somatosensory area |
| Inferior frontal gyrus | Inferior parietal area |
| Supplementary motor area | |

Mirroring movement

Some scientists suggest that mirror neurons may play a role in learning how to imitate movement. In this theory, information on the purpose of an action is passed to mirror neurons from brain areas such as the prefrontal cortex, which is responsible for analysis. Mirror neurons in various motor areas then encode a simulation of that action, which becomes part of our own motor programming. We can then go on to use this "program" if we need to carry out the action ourselves.

Observing an action

Mirror neurons respond differently to various actions of the face and limbs. In particular, neurons in different brain areas are activated for movements of the body itself, such as chewing, and those focused on a visible object, such as biting a fruit.

